

Data Collection, Cleaning & Exploratory Data Analysis (EDA)

This report demonstrates the process of collecting, cleaning, and analyzing a real-world dataset before training machine learning models. The Iris dataset was used for this task.

Dataset Used: Iris Dataset
Total Rows: 150
Total Columns: 5
Features: sepal length (cm), sepal width (cm), petal length (cm), petal width (cm)
Target Variable: target (Flower Species)

Missing Values Analysis

Column	Missing Values
sepal length (cm)	0
sepal width (cm)	0
petal length (cm)	0
petal width (cm)	0
target	0

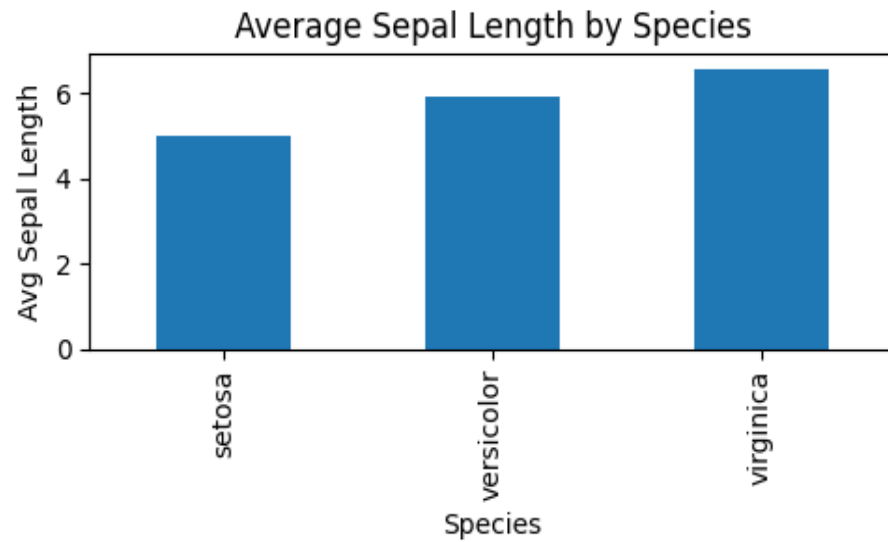
Summary Statistics

sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
150.0	150.0	150.0	150.0
5.84	3.06	3.76	1.2
0.83	0.44	1.77	0.76
4.3	2.0	1.0	0.1
5.1	2.8	1.6	0.3
5.8	3.0	4.35	1.3
6.4	3.3	5.1	1.8
7.9	4.4	6.9	2.5

- Data Cleaning Steps Performed:**
1. Checked for missing values.
 2. Verified data types of all columns.
 3. Removed duplicate rows if any.
 4. Standardized the target labels for readability.

EDA Insights:

- Setosa flowers generally have smaller petal lengths.
- Virginica flowers show larger dimensions overall.
- No missing values were found in the dataset.
- Feature distributions show clear separation between flower species.

Visualization**Conclusion:**

The dataset was successfully cleaned and analyzed using exploratory data analysis techniques. The EDA process helped identify patterns, relationships, and feature distributions, making the dataset ready for machine learning model development.